Controlled One-Click Benchmarking and Post-Fusion Recalibration for Support-Conditioned Medical Segmentation

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Keywords: Interactive segmentation; Medical image segmentation; One- click segmentation; Ultrasound segmentation; Prototype conditioning; Recalibration; Large-kernel convolution
Submission Type: Regular Track (Peer-Reviewed Workshop Paper)
TL;DR: We improve one-click prototype-conditioned medical image segmentation by introducing stage-aware recalibration that stabilizes early noisy features and late support-query fusion.

Abstract:
Sparse-prompt medical segmentation is hard to compare when prompts, supports, thresholds, and splits differ. This paper uses a controlled one-click protocol: each test image receives one positive foreground click, training-only support, validation-only checkpoint selection, and no test-time threshold tuning. We propose {Support-Aware Recalibration (SAR)}, a late post-fusion recalibration block for MultiverSeg that combines large-kernel spatial gating with squeeze-and-excitation channel recalibration after support--query fusion. BUSI, NerveUS-BP-present, and Kvasir-SEG are evaluated with matched 697 foreground-present sample budgets, shared splits/clicks, and threshold-free Dice-AUC/IoU-AUC. Compared with a protocol-matched raw-support MultiverSeg baseline, the main SAR configuration shows favorable mean Dice-AUC/IoU-AUC performance on BUSI, NerveUS-BP-present, and Kvasir-SEG, while reducing logged training time in this implementation. These controlled results indicate that late post-fusion recalibration is useful under a leakage-controlled one-click protocol.

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Paper Decision

Decision by Program Chairs 16 Jul 2026, 14:15 Program Chairs, Reviewers, Authors

Decision: Accept

A valuable controlled protocol paired with a modest, mostly single-dataset recalibration gain

Official Review by Reviewer m24M 13 Jul 2026, 23:15 (modified: 16 Jul 2026, 19:59) Program Chairs, Reviewers, Authors Revisions

Review:
Summary: The paper proposes a strict one-click evaluation protocol for support-conditioned medical segmentation plus SAR, a recalibration block on MultiverSeg's last two fusion stages combining a large-kernel spatial gate with SE channel reweighting. Evaluation uses threshold-free Dice-AUC/IoU-AUC over three seeds.

Strengths: The controlled protocol is the real contribution and a genuinely useful one, since fair comparison here is hard when prompts, support, thresholds, and splits all vary at once. The threshold-free AUC evaluation is a good choice, the ablation is reasonably thorough (SE-only, LK-only, kernel sizes, insertion placement), and the paper is honest about its three-seed, single-split limits.

Weaknesses: The evidence for SAR specifically is thin. On the clean comparison (A7 vs A1, SAR the only difference), gains are about half an AUC point and the interval clearly excludes zero only on BUSI. The larger A7 vs A2 gains change support representation and add SAR at once, so they can't cleanly credit the module. The best config is also dataset-dependent (A8 wins on Kvasir), and the training-time claim is a distraction, since SAR has more FLOPs, and the wall-clock win comes from CST precomputation, not the block. Scope is narrow: three 2D datasets at 256×256.

Overall, the evaluation protocol is a durable contribution; the module is a competent but modest add-on whose clearest benefit is on BUSI. Solid as a workshop paper on fair benchmarking, though the "recalibration reliably helps" claim isn't yet demonstrated.

Rating: 7: Good paper, accept
Confidence: 3: The reviewer is fairly confident that the evaluation is correct

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Official Review by Reviewer SWXS 10 Jul 2026, 00:17 (modified: 16 Jul 2026, 19:59) Program Chairs, Reviewers, Authors Revisions

Review:
1. Summary This paper studies support conditioned medical segmentation in a controlled one click setting. The authors fix the foreground click, use training only support, select checkpoints only on validation data, and evaluate mainly with threshold free Dice AUC and IoU AUC. The proposed method, SAR, adds a late post fusion recalibration block to MultiverSeg, combining spatial gating and channel recalibration. Experiments are reported on BUSI, NerveUS BP present, and Kvasir SEG.
2. Strengths (1) The paper addresses a relevant problem. Comparisons in promptable medical segmentation can be easily affected by different prompts, support choices, thresholds, and splits. The controlled protocol is a useful part of the work. (2) The paper is generally clear about its claims. I appreciate that the authors distinguish their controlled AUC results from source reported SAM family results. (3) The ablation study is reasonably complete for the proposed module. It includes no recalibration, raw support MultiverSeg, SE only, LK only, different kernel sizes, and different insertion locations. (4) The results are consistent across the three datasets. SAR gives small but positive gains over the matched no recalibration baseline and the raw support comparator. (5) The paper is honest about limitations, including the use of only three seeds, one fixed split, and non controlled external baseline comparisons.
3. Weaknesses (1) The main method is somewhat incremental. SAR mainly combines known spatial gating and channel recalibration after support query fusion. This is reasonable, but the novelty is limited. (2) The empirical gains are modest. The improvements over the no recalibration baseline are generally around one AUC point or less. This supports the usefulness of the module, but the evidence is not very strong. (3) The benchmark scope is still limited. The experiments use three datasets, one fixed split, and relatively small test sets. More split variation would make the robustness claim stronger. (4) The comparison with SAM family methods is not controlled. The authors state this clearly, but this table should be interpreted only as background reference rather than direct evidence of superiority. (5) The CST and raw support comparison still has some confounding, since CST is built from a 100 image training bank while the raw support baseline uses three raw supports. The paper explains this, but it limits how much can be concluded from this comparison.
4. Suggestions for improvement (1) It would be helpful to include results on additional random splits, even if only for the main baseline and SAR. (2) A controlled rerun of one strong medical SAM style baseline under the same one click protocol would strengthen the paper. (3) The authors could add a short sensitivity analysis for the sampled support bank used to build CST. (4) The discussion could be slightly expanded on cases where SAR is not clearly the best choice, for example the Kvasir result where the last block only variant is competitive.
5. Overall assessment Overall, this is a solid workshop paper. The controlled evaluation protocol is the strongest part, and the proposed recalibration module gives consistent but modest improvements. I do not view the method as a major technical advance, but the paper is relevant to the workshop and useful for discussion around fair benchmarking of promptable and support conditioned medical segmentation. I lean toward acceptance.

Rating: 7: Good paper, accept
Confidence: 3: The reviewer is fairly confident that the evaluation is correct